



S3.2 Functional groups: classification of organic compounds

Structure 3 · Classification of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Organic compounds are classified into homologous series (families with the same general formula and functional group, showing a gradual trend in physical properties and similar chemical properties) based on the functional group present, such as alkanes (C–C, C–H only), alkenes (C=C), alcohols (–OH), halogenoalkanes (–X), carboxylic acids (–COOH), esters (–COO–), aldehydes/ketones (C=O), and amines (–NH₂). IUPAC nomenclature systematically names compounds from the longest carbon chain containing the principal functional group, with substituents/locants and suffixes/prefixes indicating structure. Structural isomers have the same molecular formula but different structural arrangements (chain, positional, or functional group isomerism), often giving different physical or chemical properties.

Worked example: follow the reasoning

Draw and name all the structural isomers of the alcohol with molecular formula C₄H₁₀O that are classified as alcohols (–OH group), and classify each as primary, secondary or tertiary.

Step 1: Possible carbon skeletons for C₄ with an –OH group: straight chain (butan-) or branched chain (2-methylpropan-).

Step 2: On the straight chain, –OH can be on C1 or C2: butan-1-ol (CH₃CH₂CH₂CH₂OH, primary) and butan-2-ol (CH₃CH₂CH(OH)CH₃, secondary).

Step 3: On the branched (2-methylpropane) skeleton, –OH can be on the terminal carbon or the central carbon: 2-methylpropan-1-ol ((CH₃)₂CHCH₂OH, primary) and 2-methylpropan-2-ol ((CH₃)₃COH, tertiary).

Answer: Four isomers — butan-1-ol (primary), butan-2-ol (secondary), 2-methylpropan-1-ol (primary), 2-methylpropan-2-ol (tertiary).

Quick check · multiple choice

Which functional group is present in an ester?

- A –COOH
- B –COO–
- C –OH
- D –CHO

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Compound Q has the molecular formula C₃H₆O. [6]
 - (a) Draw the structures of two isomers of Q that belong to different homologous series (one a ketone and one an aldehyde, or an aldehyde and a cyclic/unsaturated alcohol), and name each. [3]
 - (b) State the type of isomerism shown between your two structures in (a). [1]
 - (c) One isomer of Q, propanal, can be oxidised. State the type of reagent/conditions needed and the type of organic product formed, and give its name. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) Propanal, $\text{CH}_3\text{CH}_2\text{CHO}$ (an aldehyde), and propanone, CH_3COCH_3 (a ketone) — both $\text{C}_3\text{H}_6\text{O}$. [3: two valid structures with correct names]
- (b) Functional group isomerism (same molecular formula, different functional groups/homologous series). [1]
- (c) Propanal can be oxidised using an oxidising agent such as acidified potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ (with heating, possibly under reflux for full oxidation), forming the carboxylic acid propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$. [2]

Quick check answer

B. The ester functional group is $-\text{COO}-$ (a carbonyl carbon bonded to an $-\text{O}-$ linking to the second carbon chain), formed from condensation of a carboxylic acid and an alcohol.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.